

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, Suite 101
San Luis Obispo, California**

ORDER NO. R3-2023-0010

**AMENDING THE MONITORING AND REPORTING PROGRAM
FOR
WASTE DISCHARGE REQUIREMENTS ORDER NO. R3-2020-0005
NPDES NO. CA0047856**

**NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM PERMIT FOR THE
CALIFORNIA DEPARTMENT OF CORRECTIONS AND REHABILITATION
CALIFORNIA MEN'S COLONY WASTEWATER TREATMENT PLANT**

The California Regional Water Quality Control Board, Central Coast Region (Central Coast Water Board) finds:

1. The California Department of Corrections and Rehabilitation (Discharger) is the owner and operator of the California Men's Colony wastewater treatment plant (Facility), which discharges treated wastewater to Chorro Creek, a water of the United States.
2. The Facility provides wastewater treatment for the California Men's Colony, California Army National Guard's Camp San Luis Obispo, Cuesta College, and several San Luis Obispo County facilities including the county jail, education center, and operations facility.
3. The Central Coast Water Board regulates the Facility pursuant to Waste Discharge Requirements Order No. R3-2020-0005, National Pollutant Discharge Elimination System (NPDES) Permit No. CA0047856 (Order No. R3-2020-0005 or Permit), which authorizes the Discharger to discharge tertiary treated effluent to Chorro Creek.
4. Chorro Creek recharges the Chorro Valley groundwater basin, which is a water source for the City of Morro Bay, as well as a fresh water source for the Morro Bay estuary.

Discharges of Plastic Pieces

5. On April 28, 2022, Central Coast Water Board staff inspected the Facility and observed and collected, via a 500-micron metal sieve, small plastic pieces suspended in the Facility's effluent and receiving water (Chorro Creek).

6. During an inspection of the facility on April 28, 2022, Central Coast Water Board staff observed that the Facility was discharging pieces of plastics in concentrations that pose potential adverse impacts to beneficial uses in receiving water. Due to the characteristics of plastic, the plastic pieces discharged to receiving water will likely remain in the environment for a long time. The effects of plastics in surface water include, but are not limited to, the transport of harmful pollutants, chemical leaching,¹ and obstruction of digestive tracts of aquatic life.²
7. Beneficial uses of Chorro Creek potentially impacted or adversely affected by the Facility's discharge of plastics include the following: water contact recreation (REC-1); wildlife habitat (WILD); cold fresh water habitat (COLD); warm fresh water habitat (WARM); migration of aquatic organisms (MIGR); spawning, reproduction, and/or early development (SPWN); preservation of biological habitats of special significance (BIOL); rare, threatened, or endangered species (RARE); and commercial and sport fishing (COMM). The Facility's discharge of plastics potentially also impacted beneficial uses for Morro Bay National Estuary and the Pacific Ocean at Morro Bay.³
8. On May 5, 2022, Central Coast Water Board and State Water Resources Control Board (State Water Board) Office of Enforcement staff met with the Discharger to discuss Central Coast Water Board staff's observations from its April 28, 2022 Facility inspection specific to plastics in the Facility's effluent and receiving water. During that meeting, Central Coast Water Board staff stressed the need for the Discharger to cease the discharge of plastics immediately and requested information on the Discharger's next steps to address the unauthorized discharges of plastics from the Facility.
9. On May 11, 2022, the Discharger met with a consultant at the Facility to evaluate short-term measures that the Discharger could implement to address the plastics in the Facility's effluent. On May 13, 2022, the Discharger summarized the interim measures implemented, which included source control at the California Men's Colony to reduce inmates' access to trash can liners, improved operational controls including more frequent cleaning of screens and grinders, plans for an added mechanical screening process at the effluent chamber, and a commitment to provide weekly updates to the Central Coast Water Board regarding the measures implemented to address the plastics in the Facility's effluent.

¹ Issac MN, Kandasubramanian B. Effect of microplastics in water and aquatic systems. *Environ Sci Pollut Res Int.* 2021 Apr;28(16):19544-19562. doi: 10.1007/s11356-021-13184-2. Epub 2021 Mar 2. PMID: 33655475; PMCID: PMC7924819. <https://pubmed.ncbi.nlm.nih.gov/33655475/>

² Bucci, K., Tulio, M., & Rochman, C. M. (2020). What is known and unknown about the effects of plastic pollution: A meta-analysis and systematic review. *Ecological Applications*, 30(2), e02044. <https://pubmed.ncbi.nlm.nih.gov/31758826/>

³ For a complete list of designated beneficial uses for Chorro Creek, Morro Bay National Estuary, and the Pacific Ocean at Morro Bay, reference Tables 2-1 and 2-2 in the 2019 Water Quality Control Plan for the Central Coast Basin (Basin Plan) at the following website: https://www.waterboards.ca.gov/centralcoast/publications_forms/publications/basin_plan/docs/2019_basin_plan_r3_complete_webaccess.pdf

10. On June 30, 2022, the Central Coast Water Board issued a notice of violation for the unauthorized discharges of plastics from the Facility and alleged violations of Order No. R3-2020-0005, a requirement pursuant to California Water Code (Water Code) section 13267 to submit a technical report to characterize the impacts related to the discharge of plastics from the Facility to Chorro Creek, and a requirement pursuant to Water Code section 13383 to submit a monitoring plan, that at a minimum, includes a procedure and schedule for monitoring of plastics in the Facility's effluent and receiving water.
11. On July 29, 2022, the Discharger submitted a draft monitoring plan, which was developed by technical consultant GHD, for monitoring plastics in the Facility's effluent and receiving water.
12. On November 14, 2022, the Discharger submitted a technical memorandum prepared by GHD, dated November 11, 2022, estimating that the daily dry mass of plastics discharged to Chorro Creek ranges from 30 milligrams (mg) to 8,200 mg. GHD made this estimate using 10 days of sampling data provided to GHD by the Facility chemist.

Ultraviolet (UV) System Issues

13. On June 29, 2022, the Discharger notified the Central Coast Water Board through email correspondence that the UV disinfection system reactor has not consistently operated at or above 57 percent UV transmittance at 245 nm, as required by the Permit.
14. On August 31, 2022, the Discharger self-reported eight exceedances of the effluent limit for total coliform bacteria occurring between July 14, 2022, and July 23, 2022. On November 2, 2022, the Discharger self-reported four exceedances of the total coliform bacteria effluent limit occurring between September 10, 2022, and September 17, 2022. These total coliform bacteria effluent limitation exceedances could indicate potential deficiencies in UV system maintenance.

Alternate Discharge Monitoring Location

15. On October 31, 2022, the Central Coast Water Board received a request from the discharger for approval of an alternate discharge monitoring location for M-001 due to ease of sampling and staff safety concerns. Alternate discharge monitoring location M-002, located in the vault immediately downstream of the UV disinfection system, is a suitable alternate discharge monitoring location for M-001. However, plastics monitoring must occur at discharge monitoring location M-001.

Legal Basis and Authority

16. The Central Coast Water Board is a state regulatory agency with responsibility for

protecting the quality of the waters of the state within its area of jurisdiction. The Central Coast Water Board has authority to require submission of information, direct action, establish regulations, levy penalties, and bring legal action when necessary to protect water quality.

17. The monitoring and reporting program contained in the Permit is established under Water Code sections 13267 and 13383.
18. Water Code section 13267(b)(1) authorizes the Central Coast Water Board, when investigating the water quality of waters of the state within its region in connection with the review of waste discharge requirements, to require that any person who discharges waste within the Central Coast region to provide technical or monitoring program reports that the Central Coast Water Board requires. The burden, including costs, of these reports must bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports.
19. Water Code section 13383(a) authorizes the Central Coast Region Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements for any person who owns or operates a publicly owned treatment works or other treatment works treating domestic sewage that discharges into waters of the United States. Water Code section 13383(b) further authorizes the Central Coast Water Board to require such persons to establish and maintain monitoring equipment or methods, including, where appropriate, biological monitoring methods, sample effluent as prescribed, and provide other information as may be reasonably required.
20. Order No. R3-2020-0005 establishes requirements for discharges from the Facility, including discharge prohibitions, effluent limitations, receiving water limitations, and recycled water specifications described below.
21. Order No. R3-2020-0005 does not specify a requirement to monitor for plastics in the effluent or in the receiving water. Order No. R3-2020-0005 does not authorize the discharge of plastics to Chorro Creek.
22. Order No. R3-2020-0005, Provision III, contains the following discharge prohibitions:
 - Provision III.B. The discharge of any waste not specifically regulated by this Order, excluding storm water regulated by General Permit No. CAS000001 (Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities), is prohibited.
 - Provision III.D. Creation of a condition of pollution, contamination, or nuisance, as defined by section 13050 of the Water Code, is prohibited.
 - Provision III.E. The discharge shall not cause or contribute to adverse impacts to beneficial uses of water or to threatened or endangered species and their habitat.

23. Order No. R3-2020-0005, Provision IV.A.1.d. contains the following final effluent limitation at Discharge Point No. 001:
- Floating Material. Discharge of treated wastewater through Discharge Point No. 001 shall not contain floating material, including solids, liquids, foams, and scum, in concentrations that cause nuisance or adversely affect beneficial uses.
24. Order No. R3-2020-0005, Provision V.A. contains the following receiving water limitations and surface water limitations:
- Provision V.A.3. Waters shall not contain floating material, including solids, liquids, foams, and scum, in concentrations that cause nuisance or adversely affect beneficial uses.
 - Provision V.A.4. Waters shall not contain suspended material in concentrations that cause nuisance or adversely affect beneficial uses.
 - Provision V.A.5. Waters shall not contain settleable material in concentrations that result in deposition of material that causes nuisance or adversely affects beneficial uses.
25. Order No. R3-2020-0005, Provision IV.C. includes the following recycling specifications:
- Provision IV.C.3.c.iii. The UV disinfection system reactor must operate at or above 57% UV transmittance at 245 nm.
 - Provision IV.C.3.i. The Facility shall be operated in accordance with an approved operations plan, which specifies clearly the operational limits and responses required for critical alarms. The operations plan shall be submitted to and approved by the Division of Drinking Water prior to issuance of the operating permit for the UV disinfection system. A copy of the approved operations plan shall be maintained at the treatment plant and be readily available to operations personnel and regulatory agencies.
 - Provision IV.C.3.j. The UV system must be operated with a built-in automatic reliability feature that must be triggered when the system is below the target UV dose. If the measured UV dose goes below the minimum UV dose, the UV reactor in question must alarm and startup the next available UV lamp bank or reactor.
 - Provision IV.C.3.k. specifies that conditions that should shut a reactor down and divert flow include inability to meet the target dose, high flow, low UVT, or reactor failure.
26. The Central Coast Water Board is amending the monitoring and reporting program of Order No. R3-2020-0005 to require the Discharger to monitor and report plastics contained in the Facility's effluent and receiving water. The monitoring and reporting requirements for plastics is made pursuant to Water Code section 13383.
27. The Central Coast Water Board is amending the monitoring and reporting program of Order No. R3-2020-0005 to require the Discharger to submit on a monthly basis its UV system maintenance logs to the Central Coast Water Board. This reporting

requirement is necessary to determine compliance with the recycling specifications in Order No. R3-2020-0005. The monitoring and reporting requirements for submitting monthly UV system maintenance logs is made pursuant to Water Code section 13267. The estimated cost of preparing the reports is \$1,000/month. The burden of these reports, including costs, bears a reasonable relationship to need for the reports and benefits to be obtained.

Any aggrieved person may petition the State Water Board to review the decision of the Central Coast Water Board regarding this order amending the monitoring and reporting program for Order No. R3-2020-0005. The petition must be received by the State Water Board at the following address within 30 calendar days of the Central Coast Water Board's action:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

For instructions on how to file a petition for review, see:

http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.s.html

IT IS HEREBY ORDERED THAT:

1. Attachment E, Monitoring and Reporting Program, of Waste Discharge Requirements Order No. R3-2020-0005 for the California Department of Corrections and Rehabilitation California Men's Colony Wastewater Treatment Plant is hereby amended as shown in the attachment to this order. The amended requirements shall be effective on June 1, 2023.
2. Staff is directed to prepare and post a conformed copy of Order No. R3-2020-0005 incorporating the revisions made by this order.

Matthew T. Keeling
Water Boards
Digitally signed by Matthew T. Keeling
Date: 2023.05.09 11:11:37 -07'00'
Matthew T. Keeling, Executive Officer

Attachments

1. Amended Monitoring and Reporting Program